

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives an observation a lookup identity and a content-addressed record. Agent A passes the reference. Agent B resolves the same record and checks it before continuing.

Earth data stays in the data layer. Its reference enters the reasoning layer.

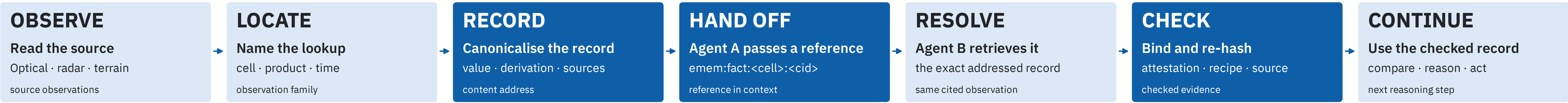
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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 How does Earth data enter an agent workflow?

Keep the source data upstream. Pass its reference.

Locate an observation, record what was read, and carry its content address into the next agent's reasoning.

EARTH DATA



Lookup identity: where / product / time

AGENT REASONING

Content address: BLAKE3 of the canonical record

CONTROLLED HANDOFF TEST: ACTED ON CORRUPTED EVIDENCE

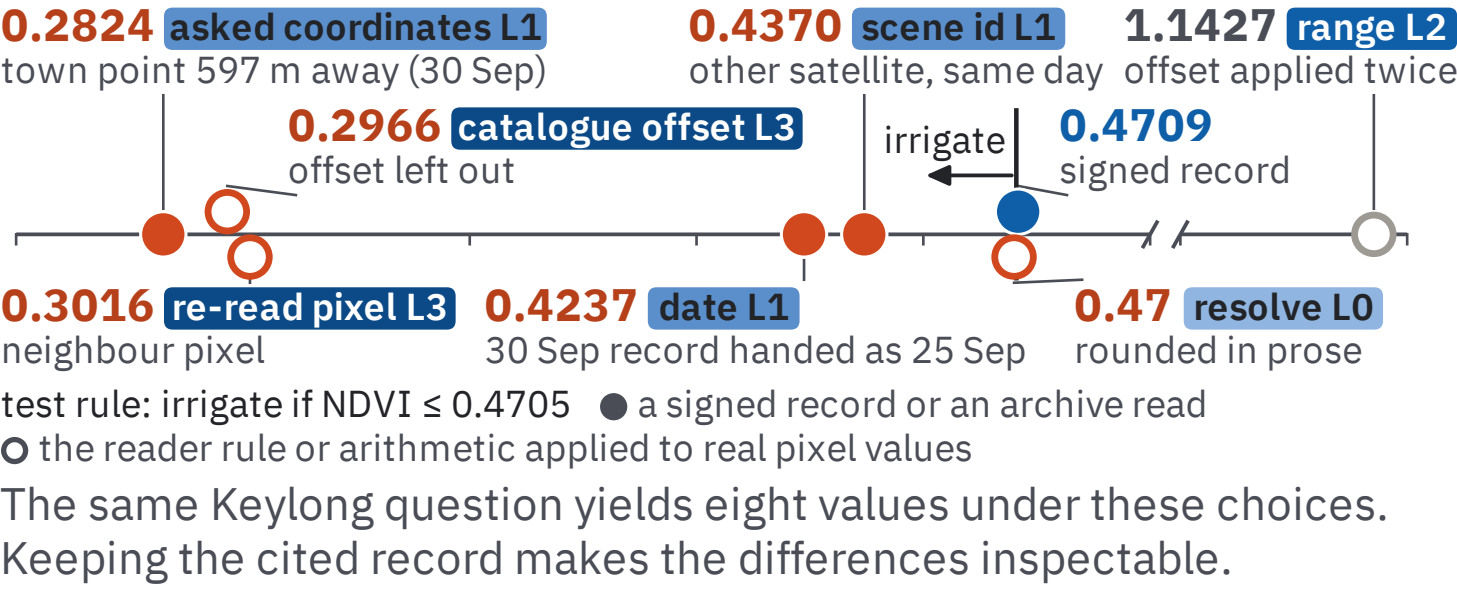
prose 254 / 276 JSON 164 / 276 retrieved text 220 / 276 opaque id 154 / 288 checked reference 0 / 300

In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

2 What can change between agents?

One question, several records

Scene, pixel, offset, rounding and stale state can change the value passed on.



3 Where did the evidence change?

Match the drift to the check

Compare the fields that identify the observation, then follow its derivation and source.

WHAT MOVED	WHAT TO CHECK
PLACE Compare the requested cell Coordinates keep the question anchored.	
TIME Bind the observation time Requested day and returned day remain visible.	
PRODUCT Inspect the band and convention Keep the product, grid and units explicit.	
SOURCE PIXEL / SCENE Re-read the named source Test the scene and the containing pixel.	
DERIVATION Recompute from the recipe Check inputs, parameters and offset rules.	
VALUE REPRESENTATION Resolve the exact record 0.47 and 0.4709 lead to different test decisions.	
REFERENT Retain the application identity The place record accompanies the object meant.	

These checks address distinct failures seen in the records and handoff tests. The methods link each example to its evidence.

Two identities, two jobs

LOOKUP IDENTITY Where · product · observation time Find the observation family.	
CONTENT ADDRESS The exact canonical record Recover what the earlier agent cited.	

emem:fact:<cell164>:<fact_cid>

The token carries the cell and record address. Product and observation time are fields inside the record.

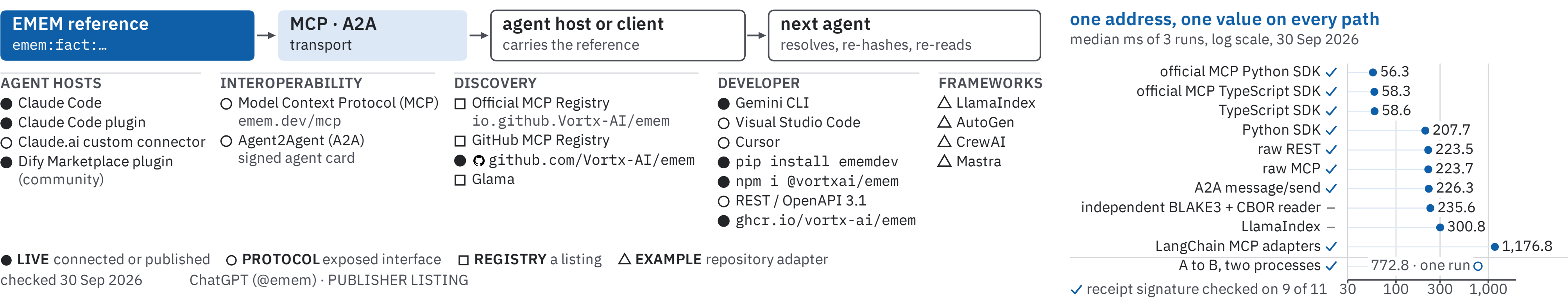
Carry one reference forward

AGENT A locates and recalls ↓ the 84-character reference ↓	
AGENT B resolves and recomputes	

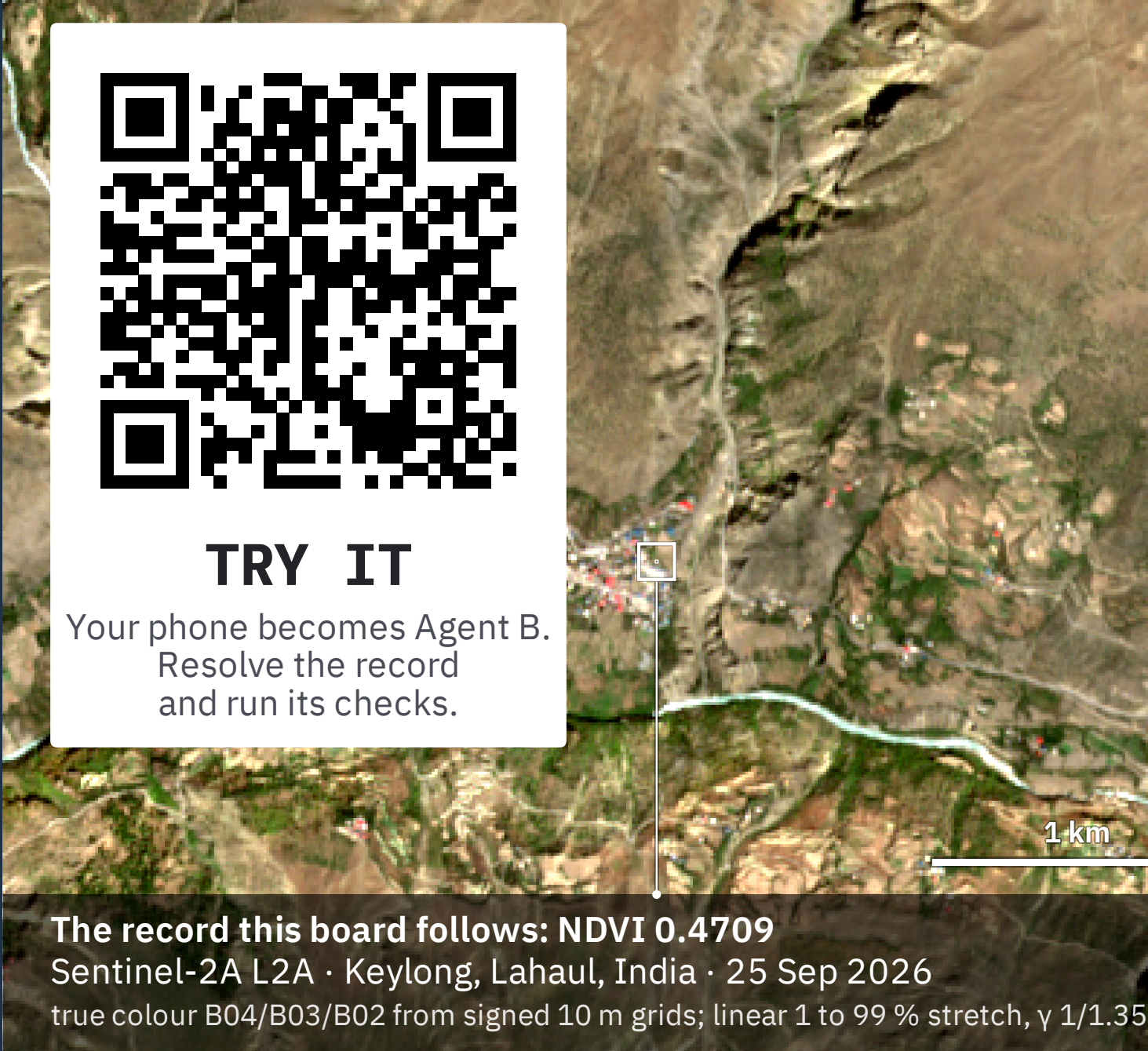
In the recorded two-process handoff, the reference alone crossed the process boundary. The receiving program re-hashed the record, checked its receipt and reproduced its NDVI exactly.

11 The same reference can cross agent runtimes

MCP, A2A, REST and SDKs expose the same reference model. One reference returned the same record and value through 11 client paths; receipt signatures were checked on 9 (30 Sep 2026). Surface status is labelled below.



Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066) · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



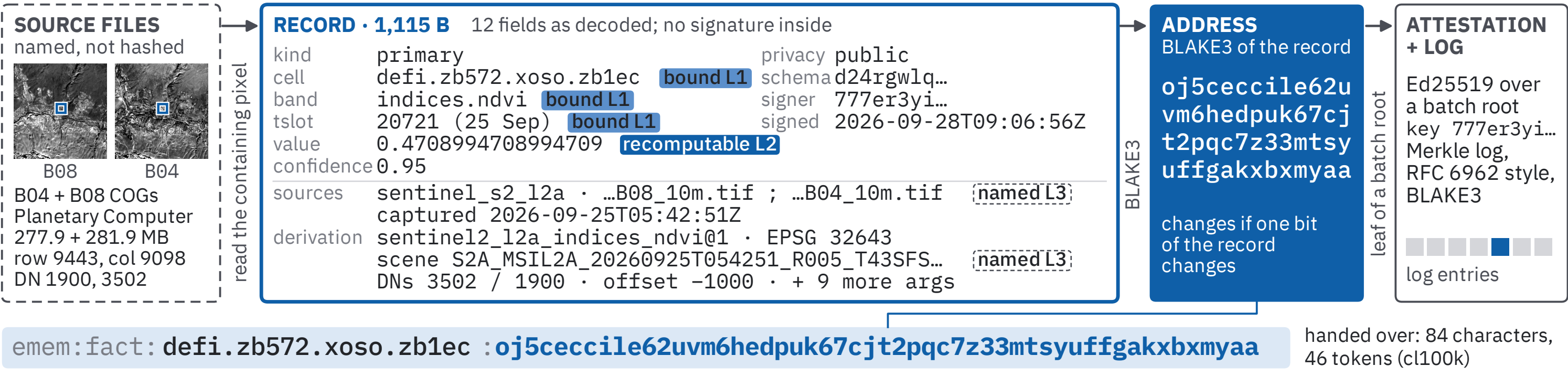
The record this board follows: NDVI 0.4709
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35

4 What exactly is handed over?

A compact reference opens the observation record

Resolve the value, place, time, derivation and source references from the record's address.

INSPECT The 1,115 bytes behind this example.



BLAKE3 identifies the exact 1,115-byte record. A batch attestation covers its address. The record names the source files and read location; source checking continues by re-reading them.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read

agents as B: three Claude models pooled, 1 Oct 2026

	prose	JSON	RAG	opaque id	EMEM, full check	first check that refuses
B acts on corrupted evidence	254 / 276	164 / 276	220 / 276	154 / 288	0 / 300	
value	M1: stated value moved by 1 ULP M2: stated value rounded to "0.47" M8: value 0.45, re-encoded and re-hashed	12/12 6/12 12/12 6/12 11/12 0/12 12/12 6/12 12/12 6/12	12/12 6/12 12/12 6/12 12/12 6/12	9/12 0/12 6/12 6/12 6/12 6/12		C L0 C L0 F L0
cell	M4: record cited for another cell M9: cell changed inside the record, re-hashed	7/12 0/12 7/12 11/12 6/12 6/12	6/12 6/12 6/12 6/12	6/12 6/12	E F	E L1 F L0
time	M10: tslot changed to look current, re-hashed	12/12 6/12 12/12 6/12 6/12 6/12	6/12 6/12 6/12 6/12	6/12 6/12	F	F L0
band	M6: a record for another band handed over	12/12 0/12 12/12 6/12 2/12 6/12	6/12 6/12	6/12 6/12	E	E L1
source	M11: source scene id changed, re-hashed	12/12 12/12 12/12 12/12 12/12 12/12	12/12 12/12	12/12 12/12	F	F L0
derivation	M12: offset changed, value recomputed, re-hashed M14: value (0.46) disagrees with the signed DNs	12/12 11/12 12/12 6/12 12/12 6/12	12/12 8/12 12/12 6/12	8/12 6/12	F H	F L0 H L2
stale / current	M5: an older record handed over as current M16: a second signed version shown only to B	12/12 1/12 12/12 6/12 6/12 6/12	6/12 1/12 6/12 6/12	1/12 6/12	E G	E L1 G L0
signature / id	M3: 1 ULP changed inside the served bytes M7: token miscopied by one character M13: value 0.45 under the forger's own key	12/12 12/12 12/12 6/12 12/12 6/12	12/12 0/12 12/12 6/12	0/12 6/12	D F	D L0 F L0
source pixel	M15: DNs read from the pixel 10 m south	12/12 12/12 12/12 12/12 12/12 12/12	12/12 12/12	12/12 12/12	I	I L3
entity	M17: same record; A meant another entity (out of scope)	not run not run not run not run	not run not run	not run not run		none

B acts on corrupted evidence

resolve L0 hash L0 binding L1 signature L0 log L0 recompute L2 re-read L3 D to I are cumulative: each adds one check
Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5, 1 Qwen 2.5-72B); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026.
Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

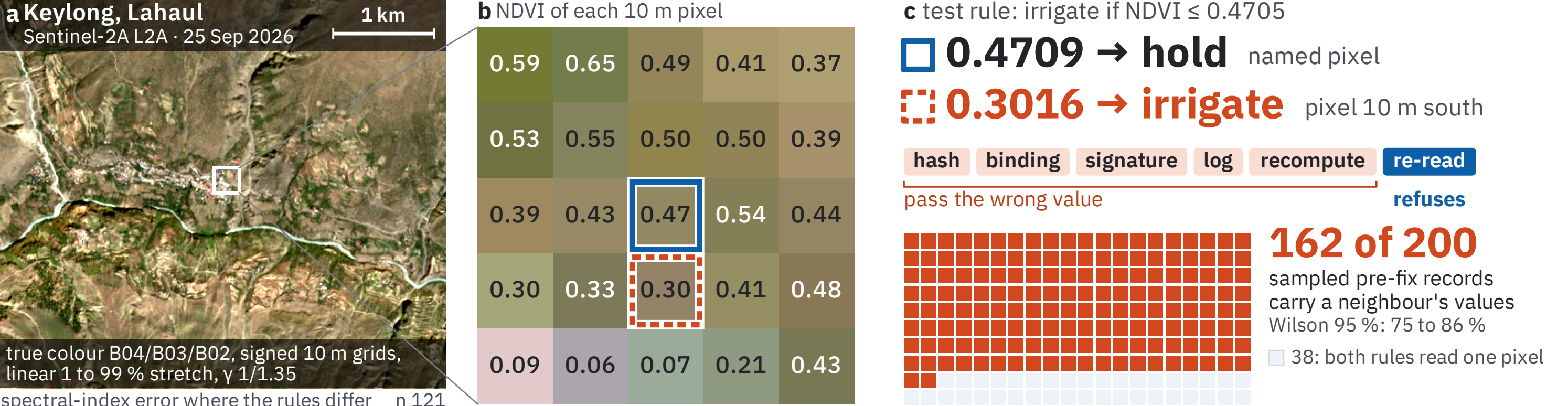
Each step answers a different question. The matrix measures which changes become visible at each depth; signatures and log checks establish attestation and publication history.

Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 What does returning to the source add?

A valid record can preserve a wrong source read

Re-reading the named source lets the receiver test the measurement itself.



the real record kxjvfwpa... 23 Sep 2026 · signed 0.3444 its containing pixel 0.4860
Sample: 200 pre-fix Sentinel-2 records cited in emem.dev's public channel, one per cell, seeded; 192 Element84, 8 Planeta Computer. After the fix, 0 of 54 (Planetary Computer, two days). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.
A record from the earlier reader still resolves: 0.3444 recorded, 0.4860 at the containing pixel. The source re-read exposes the pixel-selection difference.

7 What has the receiver established?

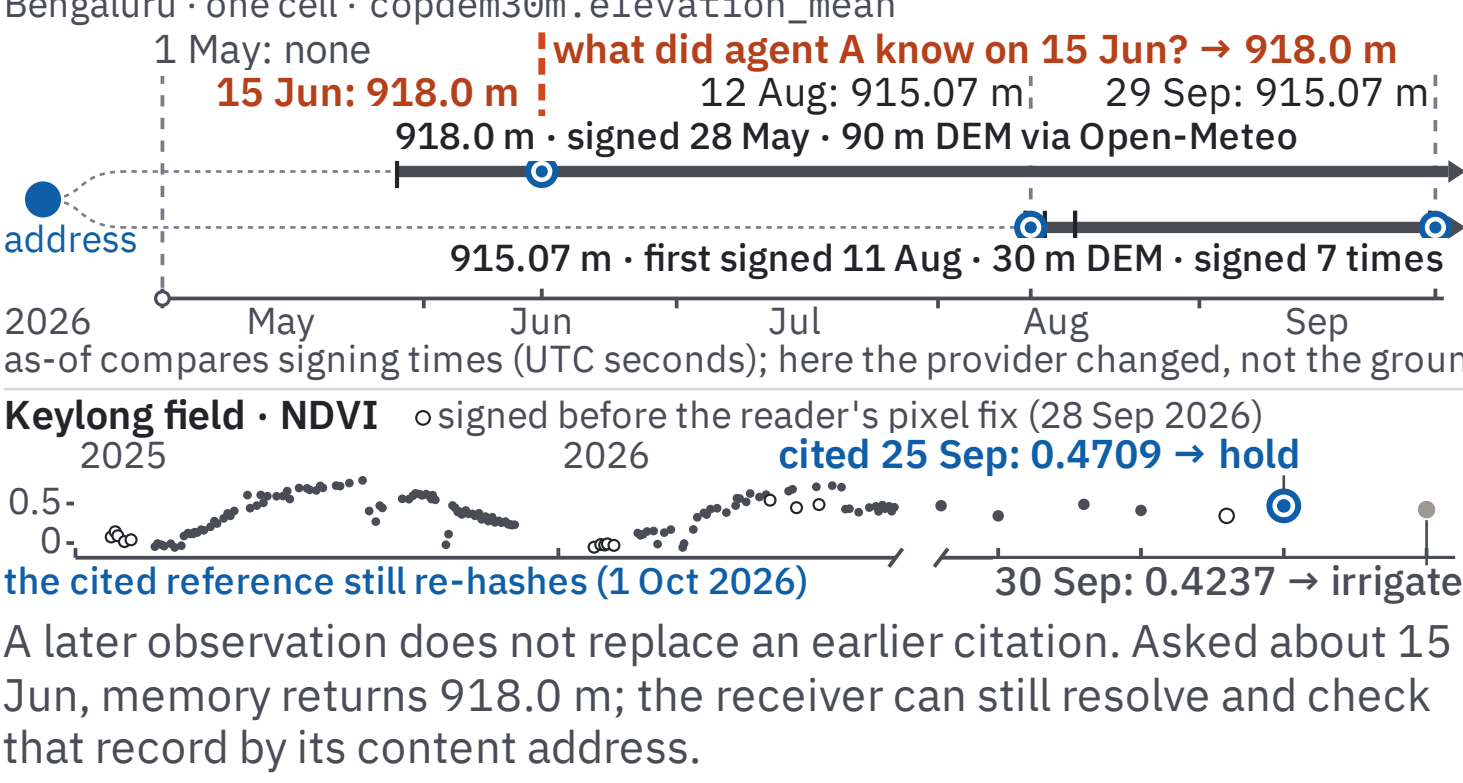
Know what was checked

RECORD	Exact addressed bytes
BINDING	Requested place, time and product
ATTESTATION	Signature under the pinned key
DERIVATION	Declared computation reproduces
SOURCE READ	Named source supports the value

Source quality remains inherited. Entity meaning and downstream decisions belong to the application using the evidence.

8 Can a later agent recover the earlier state?

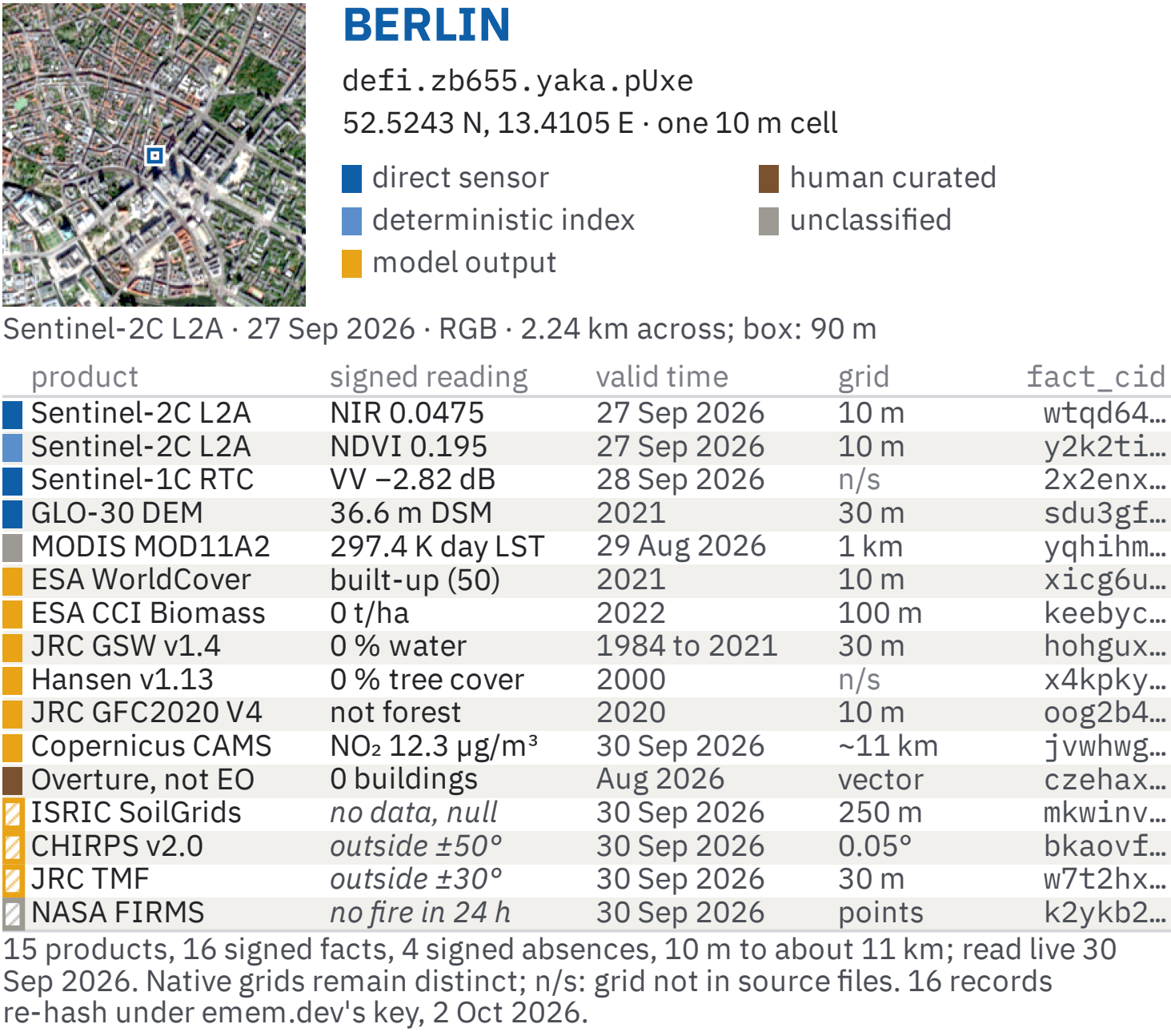
Memory keeps what was cited



9 What does one Berlin location reveal?

One place, many observations

Central Berlin: optical, radar, elevation, temperature, water, forest and biomass. One lookup cell organises 15 products, preserving each record and native grid.



10 What else can an agent carry?

Beyond a single observation

OBSERVATIONS fact · derived value · absence Recover a value and its derivation.	
EARTH DATA raster · cube · rasterset Recover a field or a series of fields.	
COLLECTIONS bundle · tree Group facts or address file chunks.	
IDENTITY AND PROCESS entity · state · trace · track Name an object, a reasoning stage or a run.	

Choose the reference that preserves what the next agent needs to recover.

EXTENSION · REFERENCE HARNESS

Bind an output to its run

SAT-042 records a test execution trace. Its gate checks the output's value digest; code and model identity are recorded in the trace. Full harness and scope in Reproduce.

12 EO data into agent reasoning

STAC describes assets; openEO defines processing; PROV and C2PA carry provenance. EMEM adds the observation reference an agent can pass on and resolve again.

JUDGE	GeoGuard	a claim in text	resolve · re-hash · re-read
CARRY	MCP, A2A	a tool call, an agent card	hand off
RETRIEVE	RAG	a text chunk	
SIGN FILES	C2PA, CDSE Traceability	a file	EMEM: is this the observation the sender cited?
RECORD LINEAGE	W3C PROV	entity, activity, agent	cite
RUN	openEO	a process graph	cite
FIND	STAC	an asset (file)	

An observation stays addressable across a handoff.

The next agent can recover the cited record, inspect its provenance, recompute a declared recipe and return to its source.

REPRODUCE

Records, code, experiments and integrations.
vortex-ai.github.io/esa_poster

